UNIVERSITY OF MINDANAO TAGUM BRANCH
BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY
Department of Computing Education
Visayan Village, Tagum City, Philippines

Course Requirements
in

Fundamental of Database (IT6/L)

Design and Implementation of a Human Resource Management (HRM) Database System for a Technology Company

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Design and Implementation of a Human Resource Management (HRM) Database System for Technology Company

Background of the Study

In a technology company, managing human resources efficiently is critical due to highly specialized roles, dynamic project environments, and rapid organizational growth. Currently, many companies rely on manual processes or outdated systems, such as spreadsheets or legacy HR software, to track employee information, job roles, skills, salaries, and project assignments.

These existing processes have several limitations:

- Data Redundancy and Inconsistency: Employee records are scattered across multiple spreadsheets or systems, leading to conflicting information.
- Limited Tracking of Employee Progression: Monitoring promotions, skill development, and salary history becomes cumbersome.
- Inefficient Project Management: Assigning employees to projects, tracking roles, and managing hours manually can result in errors and misallocation of resources.
- Restricted Reporting and Analytics: Generating reports on skills distribution, staffing, or compensation trends requires manual aggregation, which is time-consuming and error-prone.

To address these issues, a Human Resource Management (HRM) database system is proposed. A relational database ensures centralized, consistent, and reliable data storage while maintaining proper relationships among employees, job roles, departments, skills, salaries, and projects. This system allows organizations to efficiently track employee skills, career progression, salary history, and project participation. It also enables better decision-making through accurate reporting and analytics, ensuring that the right skills are available for projects and that resources are properly allocated.

For technology companies, where roles such as software engineers, data scientists, DevOps engineers, QA specialists, and project managers are dynamic and highly specialized, the proposed HRM database system enhances operational efficiency, reduces human errors, and supports employee development.

General Objective

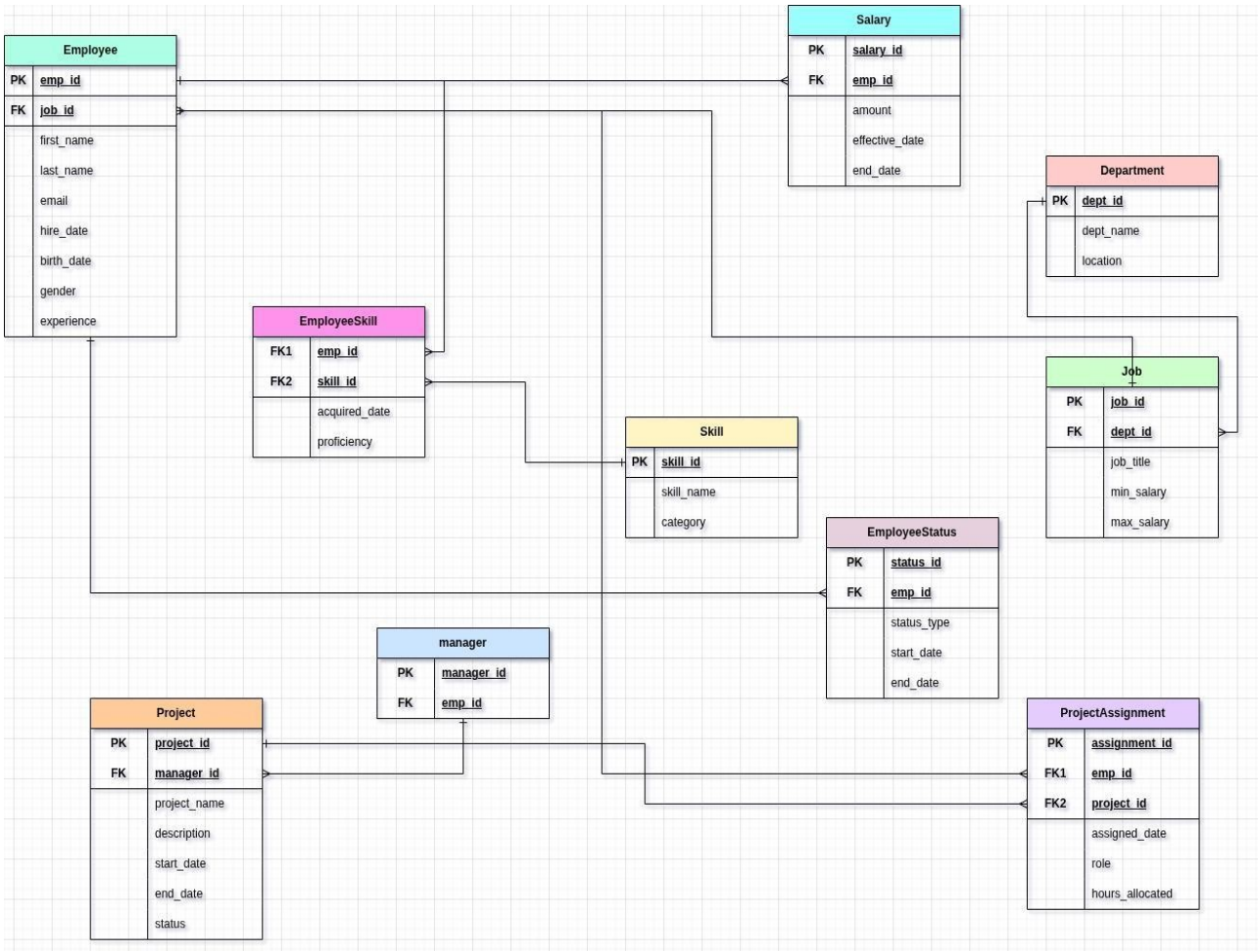
The overall goal of this study is to design and implement a comprehensive Human Resource Management (HRM) database system for a technology company that centralizes employee information, tracks skills and career progression, manages salary history, and supports project assignments and reporting.

Specific Objectives

The system aims to achieve the following measurable objectives following the **SMART principle**:

1. To create a centralized database that accurately stores employee details, job roles, departments, skills, and project assignments.
2. To track employee career progression by maintaining historical records of salary changes, promotions, and employment status.
3. To manage employee skills and proficiency levels, enabling HR to identify skill gaps and effectively plan training programs.
4. To monitor project assignments and resource allocation, ensuring that employees are assigned to appropriate projects with clearly defined roles and tracked working hours.
5. To generate accurate and timely reports for HR management, including employee statistics, salary trends, and project staffing information.
6. To reduce errors and redundancy in HR processes by enforcing relational data integrity through proper foreign key constraints.

Propose Database Design



Date Dictionary

TABLE 1: Department

Column		Data Type	Nullable	Description
dept_id	PK	INT	NOT NULL	Unique department identifier.4-digit format starting at 1001.
dept_name		VARCHAR(100)	NOT NULL	Full name of the department. E.g. “Software Engineering”, “Data Science & Analytics”
location		VARCHAR(100)	NOT NULL	City and state where the department is physically located. E.g. “San Francisco, CA”

TABLE 2: Job

Column		Data Type	Nullable	Description
job_id	PK	INT	NOT NULL	Unique department identifier.4-digit format starting at 2001.
job_title		VARCHAR(100)	NOT NULL	Official title of the role. E.g. “Senior Software Engineer”, “Data Scientist”
min_salary		DECIMAL(10,2)	NOT NULL	Minimum annual salary in USD. Defines the lower bound of the pay band.
max_salary		DECIMAL(10,2)	NOT NULL	Maximum annual salary in USD. Defines the upper bound of the pay band.
dept_id	FK	INT	NOT NULL	Department this job belongs to. References Department.dept_id (1001-1020)

TABLE 3: Employee

Column		Data Type	Nullable	Description
emp_id	PK	INT	NOT NULL	Unique department identifier.4-digit format starting at 3001.
first_name		VARCHAR(50)	NOT NULL	Employee’s given (first) name.
last_name		VARCHAR(50)	NOT NULL	Employee’s family (last) name.
email		VARCHAR(150)	NOT NULL	Company email address. Format: firstname.lastname@techcompany.com Guaranteed unique across all rows.
hire_date		DATE	NOT NULL	Date the employee officially joined the company. Range: 2015-2024.
birth_date		DATE	NOT NULL	Employee’s date of birth. Used to calculate age. Employees are 22-45 years old at time of hire.
gender		CHAR(1)	NOT NULL	Employee’s gender. M = Male, F = Female
job_id	FK	INT	NOT NULL	Current job role of the employee. References Job.job_id (2001-2024)
experience		INT	NOT NULL	Total years of professional experience at the time of hire. Range: 0-20 years.

TABLE 4: Skill

Column	Data Type	Nullable	Description
skill_id PK	INT	NOT NULL	Unique skill identifier. 4-digit format starting at 4001.
skill_name	VARCHAR(100)	NOT NULL	Name of the skill. E.g. “Python”, Kubernetes”, “Machine Learning”, “Figma”
category	VARCHAR(50)	NOT NULL	Skill category grouping. Values: Programming, Mobile, Frontend, Backend, Database, Cloud, DevOps, AI/ML, Analytics, Security, Design, Management, Architecture

TABLE 5: EmployeeSkill

Column	Data Type	Nullable	Description
emp_id PK/FK	INT	NOT NULL	References to the employee. Part of composite primary key. References Employee.emp_id (3001-3200)
skill_id PK/FK	INT	NOT NULL	References to the skill. Part of composite primary key. References Skill.skill_id (4001-4040)
acquired_date	DATE	NOT NULL	Date the employee acquired or assessed on the skill. On or after employee hire_date.
profeciency	VARCHAR(30)	NOT NULL	Proficiency level of the skill. Values: Beginner, Intermediate, Advanced, Expert

TABLE 6: Salary

Column	Data Type	Nullable	Description
salary_id PK	INT	NOT NULL	Unique department identifier.4-digit format starting at 5001.
emp_id FK	INT	NOT NULL	Employee this salary record belongs to. References Employee.emp_id(3001-3200)
amount	DECIMAL(10,2)	NOT NULL	Annual gross salary in USD for this period. Falls within the job’s min_salary - max_salary.
effective_date	DATE	NOT NULL	Date this salary amount became effectiv. Equals hire_date for initial record.
end_date	DATE	NOT NULL	Date this salary period ended. NULL = currently active salary. Set to 1 day before next effective_date on promotion.

TABLE 7: EmployeeStatus

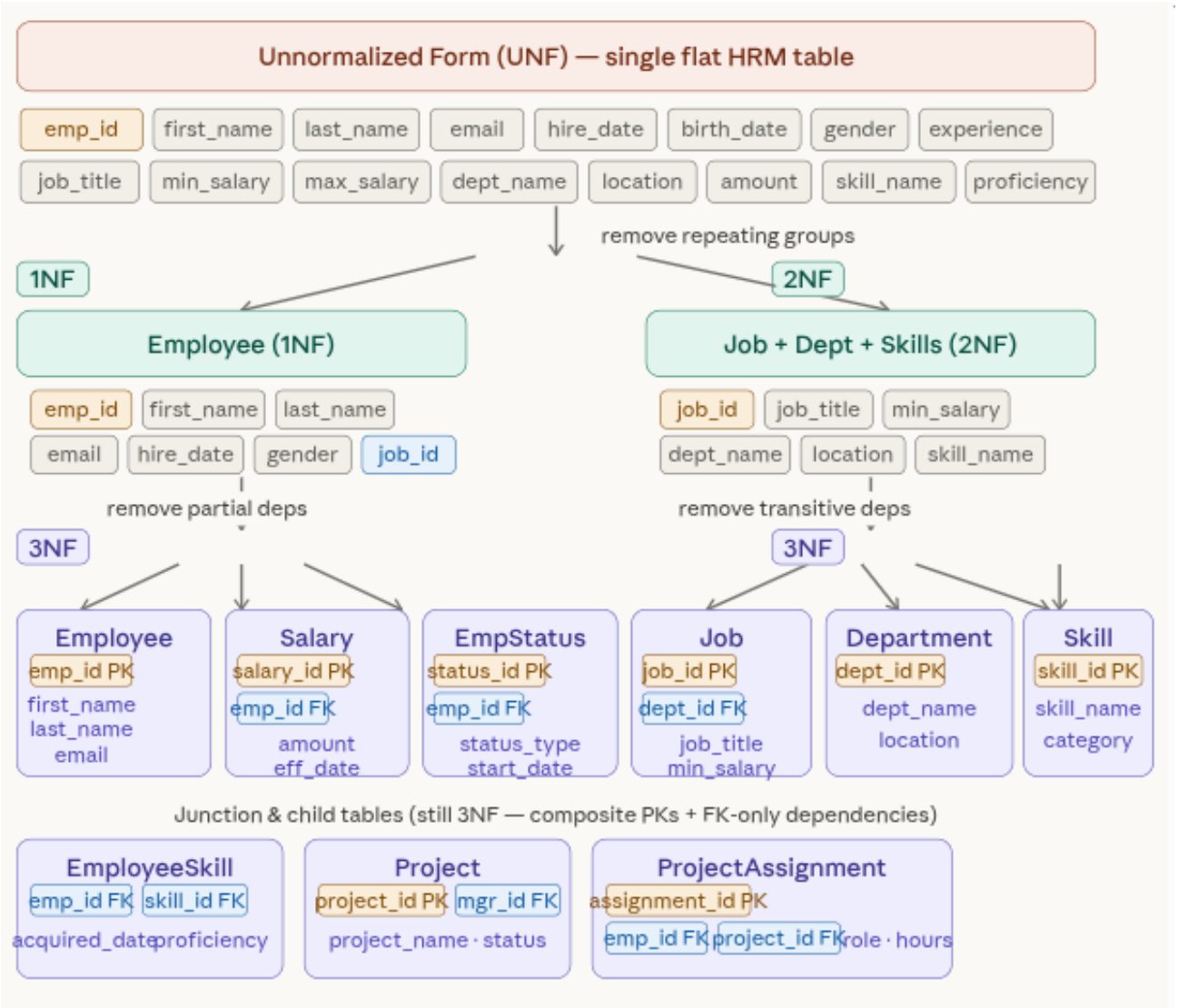
Column		Data Type	Nullable	Description
status_id	PK	INT	NOT NULL	Unique department identifier.4-digit format starting at 6001.
emp_id	FK	INT	NOT NULL	Employee this status record belongs to. References Employee.emp_id(3001-3200)
status_type		VARCHAR(50)	NOT NULL	Current employment status. Values(with distribution): Active 60%, Remote 15%, On Leave 8%, Terminated 7%, Probation 5%, Contractor 5%.
start_date		DATE	NOT NULL	Date this status period began. Equals hire_date For initial record.
end_date		DATE	NOT NULL	Date this status ended. NULL means status currently active. Populated only for Terminated.

TABLE 8: Project

Column		Data Type	Nullable	Description
project_id	PK	INT	NOT NULL	Unique project identifier. 4-digit format starting at 7001
project_name		VARCHAR(150)	NOT NULL	Internal project name. Example: "Project Phoenix", "Fraud Detection System", "MLOps Framework".
description		TEXT	NULLABLE	Brief description of the project's goals and deliverables.
start_date		DATE	NOT NULL	Official project kick-off date. Range: 2015 – 2023.
end_date		DATE	NOT NULL	Planned or actual completion date. 90–900 days after start_date.
status		VARCHAR(30)	NOT NULL	Current project status. Values (with distribution): Completed 40%, Active 35%, Planning 10%, On Hold 10%, Cancelled 5%.
manager_id	FK	INT	NOT NULL	Employee managing this project. Restricted to senior-level roles. References Employee.emp_id (3001 – 3200).

TABLE 9: ProjectAssignment

Column		Data Type	Nullable	Description
assignment_id	PK	INT	NOT NULL	Unique assignment identifier. 4-digit format starting at 8001.

emp_id	FK	INT	NOT NULL	Employee assigned to project. References Employee.emp_id (3001 – 3200).
project_id	FK	INT	NOT NULL	Project the employee is assigned to. References Project.project_id (7001 – 7050).
assigned_date		DATE	NOT NULL	Date employee was assigned. Within 30 days of project start_date.
role		VARCHAR(80)	NOT NULL	Project-specific role. Values: Developer, Tech Lead, Data Engineer, QA Engineer, DevOps Engineer, UX Designer, Product Owner, Scrum Master, Architect, ML Engineer.
hours_allocated		INT	NOT NULL	Total hours allocated to this employee for the project. Range: 20 – 200 hours.

Normalization Process